Meeting Notes with Supervisor

Meeting Note 1: Project Initialization

Date: Week 1 (Aug 2025)
Participants: Student, Supervisor
Duration: 45 minutes

Discussion Topics

1. **Project Proposal Review**
   - Discussed the project idea: integrating cuSZp compression into vLLM framework
   - Supervisor confirmed the feasibility and relevance of the project
   - Agreed on the main objective: optimizing CPU-GPU data transfer for LLM serving
2. **Project Scope Definition**
   - Focus on KV cache swapping operations in vLLM
   - Target: reduce PCIe transfer bottleneck
   - Scope: compression pipeline for swap operations only
3. **Technical Approach**
   - Supervisor suggested analyzing vLLM codebase first
   - Recommended studying cuSZp API and performance characteristics
   - Discussed error-bounded compression requirements
4. **Timeline and Milestones**
   - Agreed on 4-phase approach
   - Phase 1: Environment analysis and baseline setup (Weeks 1-3)
   - Set up regular meeting schedule: bi-weekly meetings

Action Items

- [x] Student: Analyze vLLM codebase structure
- [x] Student: Study cuSZp documentation and API
- [x] Student: Create project documentation structure
- [x] Supervisor: Review initial project plan

Next Meeting

Scheduled: Week 3 (End of Phase 1)

Meeting Note 2: Design Review

Date: Week 3 (Sept 2025)
Participants: Student, Supervisor
Duration: 50 minutes

Discussion Topics

1. **Phase 1 Completion Review**
    - Student presented analysis of vLLM's swap operations
    - Identified key functions: `swap_out_blocks_to_host` and `swap_in_blocks_from_host`
    - Discussed PagedAttention memory management mechanism
2. **Architecture Design**
    - Reviewed proposed layered architecture:
        - vLLM → CompressedSwapManager → CuSZpWrapper → cuSZp
    - Supervisor approved the modular design approach
    - Discussed Python-C++ integration strategy using pybind11
3. **Technical Decisions**
    - Agreed on using CUDA streams for asynchronous execution
    - Decided on error handling: automatic fallback mechanism
    - Confirmed compression configuration parameters
4. **Implementation Plan**
    - Start with C++ wrapper development
    - Create Python bindings
    - Then implement compression pipeline

Action Items

- [x] Student: Design C++ wrapper interface
- [x] Student: Create CMake build system
- [x] Student: Implement basic compression wrapper
- [x] Supervisor: Review wrapper design document

Next Meeting

Scheduled: Week 6 (End of Phase 2)

Meeting Note 3: Integration Progress

Date: Week 6 (Oct 2025)
Participants: Student, Supervisor
Duration: 55 minutes

Discussion Topics

1. **Phase 2 Completion Review**
   - Student demonstrated C++ wrapper implementation
   - Reviewed Python bindings using pybind11
   - Discussed PyTorch tensor integration
2. **Compression Pipeline Design**
   - Reviewed CompressedSwapManager class design
   - Discussed D2H and H2D compressed swap workflows
   - Supervisor suggested adding more detailed error logging
3. **Asynchronous Execution**
   - Reviewed CUDA stream implementation
   - Discussed overlap optimization strategies
   - Agreed on separate streams for compression and decompression
4. **Build System**
   - Reviewed CMake configuration
   - Discussed dependency management
   - Agreed on creating automated build script

Action Items

- [x] Student: Complete compression pipeline implementation
- [x] Student: Add comprehensive error logging
- [x] Student: Create automated build script (build.sh)
- [x] Student: Write integration guide documentation
- [x] Supervisor: Review compression pipeline code

Next Meeting

Scheduled: Week 11 (End of Phase 3)

Meeting Note 4: Implementation Review

Date: Week 11 (Nov 2025)
Participants: Student, Supervisor
Duration: 60 minutes

Discussion Topics

1. **Phase 3 Completion Review**
   - Student presented completed compression pipeline
   - Demonstrated CompressedSwapManager functionality

- Reviewed error handling and fallback mechanisms

2. **Code Quality**
   - Supervisor reviewed code structure
   - Discussed code documentation and comments
   - Agreed on code organization and modularity
3. **Benchmarking Framework**
   - Reviewed baseline profiling script
   - Discussed compression benchmark design
   - Agreed on performance metrics to measure
4. **Testing Strategy**
   - Discussed unit testing approach
   - Planned integration testing with vLLM
   - Identified need for GPU environment access
5. **Documentation**
   - Reviewed integration guide
   - Discussed deployment documentation
   - Agreed on adding more examples

Action Items

- [x] Student: Complete benchmarking scripts
- [x] Student: Enhance documentation with examples
- [x] Student: Create Docker deployment configuration
- [x] Student: Prepare for GPU environment testing
- [x] Supervisor: Help arrange GPU server access

Next Meeting

Scheduled: Week 15 (End of Phase 4)

Meeting Note 5: Testing and Evaluation Planning

Date: Week 15 (Dec 2025)
Participants: Student, Supervisor
Duration: 50 minutes

Discussion Topics

1. **Phase 4 Progress Review**
   - Student presented completed benchmarking framework

- Reviewed baseline profiling and compression benchmark scripts
    - Discussed Docker deployment setup
2. **Current Status**
    - Code framework: 100% complete
    - Documentation: Complete
    - Testing: Pending GPU environment access
3. **GPU Environment Access**
    - Supervisor confirmed GPU server availability
    - Discussed testing timeline
    - Planned testing schedule for next 2-3 weeks
4. **Evaluation Plan**
    - Agreed on performance metrics:
        - Transfer time reduction
        - Compression ratio
        - Overall throughput improvement
        - Accuracy impact assessment
    - Discussed test scenarios and workloads
5. **Next Steps**
    - Immediate: Compile and test in GPU environment
    - Short-term: Run baseline and compression benchmarks
    - Medium-term: Integrate with vLLM and run end-to-end tests
    - Long-term: Performance optimization and parameter tuning
6. **Interim Report**
    - Discussed interim report structure
    - Agreed on sections to include
    - Set deadline for report submission

Action Items

- [x] Student: Set up GPU environment and compile code
- [x] Student: Run baseline performance profiling
- [x] Student: Run compression benchmarks
- [x] Student: Complete interim report
- [x] Supervisor: Review interim report draft

Next Meeting

Scheduled: Week 18 (After initial GPU testing)

Summary of Project Progress

Completed (Weeks 1-15)

- ✅ Project design and architecture
- ✅ C++ wrapper implementation
- ✅ Python bindings (pybind11)
- ✅ Compression pipeline
- ✅ Benchmarking framework
- ✅ Documentation

In Progress

- ⏳ GPU environment testing
- ⏳ Performance evaluation
- ⏳ Interim report writing

Upcoming

- 📅 GPU testing and validation

- 📅 vLLM integration
- 📅 Performance analysis
- 📅 Final report preparation